The maximum number of stars in graphs with prescribed circumference or detour order

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Abstract

The circumference of a graph is the length of a longest cycle, and its detour order is the number of vertices in a longest path. In 2019, Győri, Salia, Tompkins, and Zamora gave a formula for the maximum number of copies of a star in a graph of order n and circumference c. This formula holds when n is sufficiently large relative to c, but may fail when n/c is small. Liu and Zhang later determined the corresponding maximum for complete multipartite graphs other than stars. In this paper, we close this gap by determining, for each $r \geq 3$, the exact maximum number of copies of the star on r vertices in a graph of order n and circumference c for all $n \geq c \geq 3$. We also determine the corresponding maximum for graphs of order n and detour order p for all $n \geq p \geq 3$.

Keywords: circumference; detour order; long cycles and paths; star

Mathematics Subject Classification (2020): 05C30, 05C35, 05C38

1 Introduction

All graphs considered in this paper are finite, simple, and undirected. For a graph G, let $V(G)$ and $E(G)$ denote its vertex set and edge set, respectively, and let $e(G) = |E(G)|$. The order of G is $|V(G)|$. The circumference $c(G)$ is the number of vertices in a longest cycle of G, with $c(G) = 0$ if G is acyclic. The detour order $p(G)$ is the number of vertices in a longest path of G.

For $v \in V(G)$, let $N_G(v)$ and $d_G(v)$ denote its neighborhood and degree, respectively. We omit the subscript when the graph is clear. Let $S_r = K_{1,r-1}$ be the star on r vertices, where $r \geq 2$. For graphs H and G, let $N(H, G)$ denote the number of subgraphs of G isomorphic to H. For vertex-disjoint graphs G_1 and G_2, their join $G_1 \vee G_2$ is obtained from $G_1 \cup G_2$ by adding all edges between $V(G_1)$ and $V(G_2)$. All unions of graphs are vertex-disjoint unless stated otherwise. Let K_s and $\overline{K}_s$ denote the complete and edgeless graphs of order s, respectively, and regard K_0 and $\overline{K}_0$ as the null graph. We write $G_1 \cong G_2$ if G_1 and G_2 are isomorphic.

Counting copies of a fixed graph under forbidden-subgraph constraints is a generalized Turán problem. Alon and Shikhelman [1] initiated a systematic study of these problems; see also the survey of Gerbner and Palmer [2]. Here we consider star counts when the lengths of cycles or paths are bounded.

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In the rest of the paper, assume $r \geq 3$. We use uppercase letters for the graph constructions and lowercase letters for their numbers of copies of S_r. For integers $s \geq 0$ and $x < s$, we use the convention $\binom{x}{s} = 0$; also, $\binom{x}{0} = 1$ for $x \geq 0$.

Notation 1. *Let $n \geq c \geq 2$ and $1 \leq k \leq \lfloor c/2 \rfloor$. Define*

$$G(n, c; k) = K_k \vee \left(K_{c+1-2k} \cup \overline{K}_{n-c-1+k}\right). \tag{1.1}$$

Let A, C, D denote the three vertex classes in (1.1), respectively. Thus $|A| = k$, $|C| = c + 1 - 2k$, and $|D| = n - c - 1 + k$. Every vertex of A has degree $n-1$, C induces a clique, D is an independent set, and there are no edges between C and D. Thus, we have

$$\begin{aligned} g_r(n, c; k) &:= N(S_r, G(n, c; k)) \\ &= k\binom{n-1}{r-1} + (n - c - 1 + k)\binom{k}{r-1} + (c + 1 - 2k)\binom{c-k}{r-1}. \end{aligned} \tag{1.2}$$

For convenience, write $G(n, c) = G(n, c; \lfloor c/2 \rfloor)$ and $g_r(n, c) = N(S_r, G(n, c))$.

Notation 2. *Let $n \geq c \geq 3$, and write*

$$n - 1 = \alpha(c - 1) + \beta, \qquad 0 \leq \beta \leq c - 2.$$

Let $F(n, c) = K_1 \vee (\alpha K_{c-1} \cup K_\beta)$. Then

$$f_r(n, c) := N(S_r, F(n, c)) = \binom{n-1}{r-1} + \alpha(c - 1)\binom{c-1}{r-1} + \beta\binom{\beta}{r-1}.$$

Notation 3. *Let $n \geq p \geq 3$, and write*

$$n = \alpha p + \beta, \qquad 0 \leq \beta \leq p - 1.$$

Let $H(n, p) = \alpha K_p \cup K_\beta$. Then

$$h_r(n, p) := N(S_r, H(n, p)) = \alpha p\binom{p-1}{r-1} + \beta\binom{\beta-1}{r-1}.$$

A graph of order $n \geq 3$ is Hamiltonian if and only if its circumference is n. Erdős and Gallai proved the following classical bound.

Theorem 1.1 (Erdős and Gallai [3])**.** *Let $n \geq c \geq 3$. If G is an n-vertex graph with circumference c, then*

$$e(G) \leq \frac{c(n-1)}{2}.$$

Kopylov obtained the corresponding sharp bound for 2-connected graphs.

Theorem 1.2 (Kopylov [4])**.** *Let $n - 1 \geq c \geq 4$. If G is a 2-connected graph of order n and circumference c, then*

$$e(G) \leq \max\{e(G(n, c; 2)), e(G(n, c))\}.$$

For paths, Faudree and Schelp [5] and independently Kopylov [4] determined the exact maximum number of edges in graphs without a long path. In the connected setting, Kopylov obtained the sharp bound, and Balister, Győri, Lehel, and Schelp [6] later gave a constructive proof and characterized the extremal graphs.

Luo [7] extended Kopylov's result from edges to complete graphs. Chakraborti and Chen [8] determined the exact clique maxima in graphs without long paths or cycles and characterized all extremal graphs. Lu, Yuan, and Zhang [9] studied complete bipartite subgraphs in graphs without long cycles or paths. Their result for graphs with prescribed detour order can be stated as follows.

Theorem 1.3 (Lu, Yuan, and Zhang [9]). *Let $n \geq p \geq 4$ and $b \geq a \geq 2$. If G is an n-vertex graph with detour order p, then*

$$N(K_{a,b}, G) \leq \max\{N(K_{a,b}, H(n,p)), N(K_{a,b}, G(n,p-1))\}.$$

Here $G(n,p-1) = G(n,p-1;\lfloor (p-1)/2 \rfloor)$. In the notation of [9], the forbidden path has $p+1$ vertices. The case $n = p$ follows directly from $H(p,p) = K_p$.

Liu and Zhang [10] extended the bounds for 2-connected graphs with prescribed circumference and connected graphs with prescribed detour order to complete multipartite subgraphs. Their results in these two settings include stars. For graphs without connectivity assumptions, they determined the corresponding maxima for complete multipartite graphs other than stars.

Star counts are closely related to sums of powers of vertex degrees. Caro and Yuster [11] studied extremal problems for such degree sums under forbidden-subgraph constraints, and Gerbner [12] developed the connection with star-counting problems.

For stars, Győri, Salia, Tompkins, and Zamora [13, Theorems 15 and 17] obtained exact formulas for sufficiently large order in the path and cycle problems. The cycle formula need not hold for small order relative to the circumference. For example, when $r = 3$, $n = 6$, and $c = 5$, we have $f_3(6,5) = 34 > 28 = g_3(6,5)$. Our aim is to remove the large-order assumption in both problems and determine the exact maxima for all admissible orders.

Our main results are as follows.

Theorem 1.4. *Let $r \geq 3$ and $n \geq c \geq 3$. If G is an n-vertex graph with circumference c, then*

$$N(S_r, G) \leq \max\{f_r(n,c), g_r(n,c)\}.$$

The bound is sharp, as shown by $F(n,c)$ and $G(n,c)$.

Theorem 1.5. *Let $r \geq 3$ and $n \geq p \geq 3$. If G is an n-vertex graph with detour order p, then*

$$N(S_r, G) \leq \max\{h_r(n,p), g_r(n,p-1)\}.$$

The bound is sharp, as shown by $H(n,p)$ and $G(n,p-1)$.

Both terms in each maximum are necessary, since either may be larger depending on the parameters. We do not classify all graphs attaining equality.

Section 2 contains the preliminary results. We prove Theorems 1.4 and 1.5 in Sections 3 and 4, respectively, and conclude with applications to non-Hamiltonian and non-traceable graphs.

2 Preliminaries

We first verify the circumference and detour order of the constructions introduced in Section 1.

Proposition 2.1. *The following statements hold.*

(i) $c(F(n,c)) = c$ *for* $n \geq c \geq 3$;

(ii) $p(H(n,p)) = p$ *for* $n \geq p \geq 3$;

(iii) $c(G(n,c;k)) = c$ *for* $n \geq c \geq 3$ *and* $1 \leq k \leq \lfloor c/2 \rfloor$;

(iv) $p(G(n,p-1;k)) = p$ *for* $n \geq p \geq 3$ *and* $1 \leq k \leq \lfloor (p-1)/2 \rfloor$.

Proof. Every cycle of $F(n,c)$ is contained in the subgraph induced by its central vertex and one of the remaining cliques. These cliques have order at most $c-1$, and at least one has order $c-1$. Hence $c(F(n,c)) = c$. Similarly, every component of $H(n,p)$ is a clique of order at most p, and at least one has order p. Thus $p(H(n,p)) = p$.

Now consider $G(n, c; k)$ with the partition A, C, D. A cycle or path avoiding A lies in C or consists of a single vertex of D. If it contains $a \geq 1$ vertices of A, deleting them leaves at most a components in the cycle and at most $a+1$ in the path. Each vertex of D forms a separate component, and any vertices in C occupy at least one further component. Since $a \leq k$, every cycle has at most $\max\{2k, 2k-1+|C|\} = c$ vertices, and every path has at most $\max\{2k+1, 2k+|C|\} = c+1$ vertices.

Since $|D| \geq k-1$, we can alternate all k vertices of A with $k-1$ vertices of D, starting and ending in A, then traverse all vertices of C and return to the first vertex. For $c \geq 3$, this is a cycle of length $2k-1+|C| = c$, proving (iii). If $n \geq c+1$, then $|D| \geq k$. Alternating k vertices of D with all vertices of A, starting in D, and then traversing C gives a path of order $2k+|C| = c+1$. Taking $c = p-1$ proves (iv). □

We next prove the convexity and monotonicity properties needed below. For a function φ on the nonnegative integers, let $\Delta\varphi(x) = \varphi(x+1) - \varphi(x)$. We call φ discretely convex if $\Delta\varphi$ is non-decreasing, equivalently $\Delta^2\varphi(x) = \varphi(x+2) - 2\varphi(x+1) + \varphi(x) \geq 0$.

Lemma 2.2. *Let L be a positive integer, and let $\varphi : \mathbb{Z}_{\geq 0} \to \mathbb{R}$ be discretely convex with $\varphi(0) = 0$.*

(i) *If x, y are nonnegative integers, then $\varphi(x+y) \geq \varphi(x) + \varphi(y)$.*

(ii) *If x, y are integers with $0 \leq x \leq y < L$ and $x + y > L$, then*

$$\varphi(x) + \varphi(y) \leq \varphi(x+y-L) + \varphi(L).$$

(iii) *Let N be a nonnegative integer. Among all finite sequences of nonnegative integers $a_i \leq L$ with sum N, the maximum of $\sum_i \varphi(a_i)$ is attained by*

$$\underbrace{L, \ldots, L}_{\lfloor N/L \rfloor \text{ terms}}, \quad N - L\lfloor N/L \rfloor.$$

The final term is omitted if it is zero. For a fixed sequence length, zero terms may be added.

Proof. For integers $1 \leq u \leq v$, discrete convexity gives

$$\varphi(u-1) + \varphi(v+1) - \varphi(u) - \varphi(v) = \Delta\varphi(v) - \Delta\varphi(u-1) \geq 0.$$

Thus transferring units from a smaller positive argument to a larger one does not decrease the sum. Transferring all units from the smaller of x, y proves (i), since $\varphi(0) = 0$. Under the assumptions of (ii), transferring $L - y$ units from x to y gives the desired pair $(x+y-L, L)$. For (iii), whenever two terms lie strictly between 0 and L, transfer units from the smaller to the larger until one becomes 0 or L. Each operation reduces the number of terms strictly between 0 and L. The final sequence has $\lfloor N/L \rfloor$ terms equal to L and the stated remainder, together with any zero terms needed to preserve its length. □

Lemma 2.3. *For fixed $n \geq c \geq 2$ and $r \geq 3$, the sequence $g_r(n, c; k)$ is discretely convex for $1 \leq k \leq \lfloor c/2 \rfloor$.*

Proof. For every k for which $k, k+1, k+2$ lie in the stated interval, Pascal's identity and (1.2) give

$$\begin{aligned}\Delta^2 g_r(n, c; k) = {} & 2\binom{k}{r-2} + (n-c+k+1)\binom{k}{r-3} \\ & + 4\binom{c-k-2}{r-2} + (c+1-2k)\binom{c-k-2}{r-3} \geq 0.\end{aligned}$$

All coefficients and binomial terms are nonnegative. □

Lemma 2.4. *For fixed n and $r \geq 3$, the following functions are non-decreasing on the indicated integer ranges:*

(i) $f_r(n,c)$ *for* $3 \leq c \leq n$;

(ii) $g_r(n,c)$ *for* $2 \leq c \leq n$;

(iii) $h_r(n,p)$ *for* $3 \leq p \leq n$.

Proof. For $x \geq 0$, let $f(x) = x\binom{x}{r-1}$ and $g(x) = x\binom{x-1}{r-1}$. The identities

$$f(x) = (r-1)\binom{x}{r-1} + r\binom{x}{r}, \qquad g(x) = r\binom{x}{r}$$

show that f and g are non-decreasing and discretely convex, since

$$\Delta\binom{x}{j} = \binom{x}{j-1} \quad (j \geq 1), \qquad \Delta^2\binom{x}{j} = \binom{x}{j-2} \quad (j \geq 2).$$

The stars centered at the non-central vertices of $K_1 \vee (\bigcup_i K_{a_i})$ contribute $\sum_i f(a_i)$. By Lemma 2.2, among partitions of $n-1$ with all parts at most $c-1$, this sum is maximized by α parts of size $c-1$ and one part of size β, where α, β are as in the definition of $F(n,c)$. Increasing c enlarges the set of allowed partitions, while the contribution $\binom{n-1}{r-1}$ of the central vertex is unchanged. This proves (i). Similarly, a component K_{a_i} contains $g(a_i)$ copies of S_r. By Lemma 2.2, the partition defining $H(n,p)$ maximizes $\sum_i g(a_i)$ when every part is at most p. Increasing p enlarges the set of allowed partitions, proving (iii).

For (ii), we show that $G(n,c)$ is a spanning subgraph of $G(n,c+1)$ whenever $c < n$. If $c = 2t$, choose one vertex of the independent part of $G(n,2t) = K_t \vee (K_1 \cup \overline{K}_{n-t-1})$ and join it to the K_1-vertex. The resulting graph is $G(n,2t+1) = K_t \vee (K_2 \cup \overline{K}_{n-t-2})$. If $c = 2t+1$, choose one vertex of the K_2-part of $G(n,2t+1)$ and join it to every vertex of the independent part. The chosen vertex becomes universal, and the other vertex of the K_2-part forms the new singleton clique. The resulting graph is $G(n,2t+2)$. In both cases, adding edges cannot decrease the number of stars. □

The star cases of Corollaries 15 and 16 in [10] give the following two lemmas.

Lemma 2.5 (Liu and Zhang [10])**.** *Let $r \geq 3$ and $n-1 \geq c \geq 4$. If G is a 2-connected n-vertex graph with circumference c, then*

$$N(S_r, G) \leq \max\{g_r(n,c;2), g_r(n,c)\}.$$

Lemma 2.6 (Liu and Zhang [10])**.** *Let $r \geq 3$, $n \geq p+1$, and $p \geq 4$. If G is a connected n-vertex graph with detour order p, then*

$$N(S_r, G) \leq \max\{g_r(n,p-1;1), g_r(n,p-1)\}.$$

3 Graphs with prescribed circumference

Proof of Theorem 1.4. We first transform G without changing its order or circumference and without decreasing $N(S_r, G)$. We continue to denote the resulting graph by G. Connect its components by bridges. This creates no cycle and cannot decrease any degree. We regard each bridge as a block isomorphic to K_2.

Choose a vertex v of maximum degree. Suppose that an end-block Q does not contain v, and let w be its cut vertex. Replace each edge wu, where $u \in N_Q(w)$, by vu, and let $x = d_Q(w)$. Since

the vertices of $Q - w$ have no neighbors outside Q, the resulting graph is simple. The copy of Q now meets the rest of the graph only at v, so the circumference is unchanged. Only the degrees of v and w change. In terms of the degrees before replacement, the change in the star count is

$$\begin{aligned}&\binom{d(v)+x}{r-1}+\binom{d(w)-x}{r-1}-\binom{d(v)}{r-1}-\binom{d(w)}{r-1}\\&\quad=\sum_{j=0}^{x-1}\left[\binom{d(v)+j}{r-2}-\binom{d(w)-x+j}{r-2}\right]\geq 0.\end{aligned}$$

The inequality follows from $d(v) \geq d(w)$. Thus v remains of maximum degree, and its degree strictly increases at each step. Hence the process terminates. Every block then contains v. Indeed, root the block–cut tree at v if v is a cut vertex, and otherwise at the unique block containing v. A block not containing v would lie on a branch ending in an end-block not containing v, a contradiction.

We next modify the blocks. Complete every block of order at most c to a clique. Since every cycle lies in a block and no edge is deleted, this preserves the circumference of the whole graph. Now consider a block Q of order $s \geq c+1$, and let $d = c(Q) \leq c$. A 2-connected graph of order at least four contains a cycle of length at least four, so $d \geq 4$. By Lemmas 2.5 and 2.3,

$$N(S_r, Q) \leq \max\{g_r(s, d; 1), g_r(s, d)\}.$$

Since $r \geq 3$,

$$g_r(s, d; 1) = \binom{s-1}{r-1} + (d-1)\binom{d-1}{r-1} \leq f_r(s, d).$$

Indeed, in the definition of $f_r(s, d)$, at least one part has size $d-1$ because $s - 1 \geq d$. Lemma 2.4 therefore gives

$$N(S_r, Q) \leq \max\{f_r(s, c), g_r(s, c)\}.$$

Replace Q by a graph $R \in \{F(s, c), G(s, c)\}$ with the larger star count, identifying a universal vertex of R with v. By Proposition 2.1, R has circumference c, so the whole graph still has circumference c.

We also check the stars centered at v whose leaves lie in several blocks. Let $x = d_Q(v)$, and let X be the number of neighbors of v outside Q. The change in the total star count is

$$N(S_r, R) - N(S_r, Q) + \left[\binom{X+s-1}{r-1} - \binom{s-1}{r-1}\right] - \left[\binom{X+x}{r-1} - \binom{x}{r-1}\right] \geq 0.$$

Indeed, the first difference is nonnegative and $x \leq s-1$. Moreover, $\binom{X+z}{r-1} - \binom{z}{r-1}$ is non-decreasing in z, since its forward difference is $\binom{X+z}{r-2} - \binom{z}{r-2} \geq 0$.

After these replacements, v is universal and each block is either a clique of order at most c or a graph $G(s, c)$ with $s \geq c+1$. In the latter case, $c \geq 4$, so $G(s, c)$ has at least two universal vertices and is 2-connected. For each block Q, let $N^\circ(S_r, Q)$ denote the number of stars in Q whose center is different from v. Then

$$N(S_r, G) = \binom{n-1}{r-1} + \sum_Q N^\circ(S_r, Q). \tag{3.1}$$

Suppose first that there is no G-block. Write the block orders as $a_i + 1$, where $1 \leq a_i \leq c-1$ and $\sum_i a_i = n-1$. Their total contribution to the sum in (3.1) is $\sum_i a_i \binom{a_i}{r-1}$. The function $f(a) = a\binom{a}{r-1}$ is discretely convex and satisfies $f(0) = 0$. Lemma 2.2(iii), with cap $c-1$, now gives $N(S_r, G) \leq f_r(n, c)$. This also proves the case $c = 3$, since every block then has order at most three.

It remains to consider graphs containing a G-block. Let $t = \lfloor c/2 \rfloor \geq 2$ and $\varepsilon = c+1-2t \in \{1, 2\}$. Since v belongs to the universal class,

$$N^\circ(S_r, G(s,c)) = (t-1)\binom{s-1}{r-1} + (s-c-1+t)\binom{t}{r-1} + \varepsilon\binom{c-t}{r-1}. \tag{3.2}$$

We show that all blocks can be replaced by a single G-block without decreasing the star count. Each replacement below preserves the order and circumference and keeps v universal. By (3.1), it suffices to compare the stars whose centers are different from v.

First consider two G-blocks of orders $x+1$ and $y+1$, where $x, y \geq c$. Replace them by $G(x+y+1, c)$. By (3.2), the change in their contribution is

$$(t-1)[\binom{x+y}{r-1} - \binom{x}{r-1} - \binom{y}{r-1}] + (c-t)\binom{t}{r-1} - \varepsilon\binom{c-t}{r-1}.$$

Vandermonde's identity and $r \geq 3$ give

$$\begin{aligned}\binom{x+y}{r-1} - \binom{x}{r-1} - \binom{y}{r-1} &= \sum_{i=1}^{r-2}\binom{x}{i}\binom{y}{r-1-i} \\ &\geq c\binom{c}{r-2} \geq 2\binom{c-t}{r-1}.\end{aligned}$$

The last inequality follows from $2\binom{z}{r-1} \leq z\binom{z}{r-2}$ with $z = c-t$ and monotonicity in z. Since $t-1 \geq 1$ and $\varepsilon \leq 2$, the change is nonnegative. Repeating this operation leaves a single G-block.

Next consider a block $G(x+1, c)$ with $x \geq c$ and a complete block K_{y+1} with $1 \leq y \leq c-1$. Replace them by $G(x+y+1, c)$. By (3.2), the change in their contribution is

$$(t-1)[\binom{x+y}{r-1} - \binom{x}{r-1}] + y\binom{t}{r-1} - y\binom{y}{r-1}.$$

If $r-1 > y$, then $\binom{y}{r-1} = 0$, so the change is nonnegative. Otherwise, $3 \leq r \leq y+1$, and for every $z \geq y-1$,

$$\frac{\binom{z}{r-2}}{\binom{y-1}{r-2}} = \prod_{i=0}^{r-3}\frac{z-i}{y-1-i} \geq \frac{z}{y-1}.$$

Summing the forward differences of $z \mapsto \binom{z}{r-1}$ and using $x \geq y+1$, we obtain

$$\frac{\binom{x+y}{r-1} - \binom{x}{r-1}}{\binom{y}{r-1}} \geq \frac{r-1}{y(y-1)}\sum_{j=0}^{y-1}(x+j) = \frac{(r-1)(x+(y-1)/2)}{y-1} \geq \frac{3y+1}{y-1}.$$

Since $c \geq 4$ implies $3(t-1) \geq c-2 \geq y-1$, it follows that $(t-1)[\binom{x+y}{r-1} - \binom{x}{r-1}] \geq y\binom{y}{r-1}$. Thus this replacement also does not decrease the star count. Repeating it for every complete block gives $G(n, c)$.

The two cases prove the required upper bound. By Proposition 2.1, both $F(n, c)$ and $G(n, c)$ have circumference c and attain the respective candidate values. Hence the bound is sharp. □

4 Graphs with prescribed detour order

Proof of Theorem 1.5. We first show that every connected graph Q of order q and detour order $h \geq 3$ satisfies

$$N(S_r, Q) \leq \max\{h_r(q,h), g_r(q,h-1)\}. \tag{4.1}$$

The case $q = h$ follows by adding edges to obtain K_h. Hence assume that $q > h$. If $h = 3$, let $x_1x_2x_3$ be a longest path in Q. Neither endpoint has a neighbor outside the path. Since Q is connected, every component of $Q - \{x_1, x_2, x_3\}$ has a vertex adjacent to x_2. If such a component contains an edge, a vertex adjacent to x_2 has a neighbor in the same component, yielding a four-vertex path ending with x_2x_1. Thus every such component is a single vertex adjacent only to x_2. Moreover, $x_1x_3 \notin E(Q)$, since otherwise an outside vertex followed by $x_2x_1x_3$ would form a four-vertex path. Hence $Q = K_{1,q-1} = G(q, 2)$.

Suppose that $h \geq 4$. By Lemma 2.6,

$$N(S_r, Q) \leq \max\{g_r(q, h-1; 1), g_r(q, h-1)\}.$$

The two candidates coincide when $h = 4$. For $h \geq 5$, let $t = \lfloor (h-1)/2 \rfloor \geq 2$. Since $r \geq 3$, we have $(h-1)\binom{h-2}{r-1} \leq (h-3)\binom{h-1}{r-1}$. Also, $t \geq (h-2)/2$ implies

$$t(4h-6) - h(h-1) \geq h^2 - 6h + 6 > 0.$$

Consequently, $t(h-2)(h-3) \leq (t-1)h(h-1)$, and hence

$$(h-2)\binom{h-2}{r-1} \leq \frac{(h-2)(h-3)}{h-1}\binom{h-1}{r-1} \leq \frac{t-1}{t}h\binom{h-1}{r-1}.$$

By (1.2), it follows that

$$\begin{aligned} g_r(q, h-1; 1) &= \binom{q-1}{r-1} + (h-2)\binom{h-2}{r-1} \\ &\leq \frac{1}{t}\left[t\binom{q-1}{r-1}\right] + \frac{t-1}{t}\left[h\binom{h-1}{r-1}\right] \\ &\leq \max\{g_r(q, h-1), h_r(q, h)\}. \end{aligned}$$

The last inequality follows from $g_r(q, h-1) \geq t\binom{q-1}{r-1}$ and $h_r(q, h) \geq h\binom{h-1}{r-1}$, since $H(q, h)$ contains a component K_h. This proves (4.1).

We now apply this bound to the components of G. Add edges to make every component of order at most p complete. If a component has order $q > p$ and detour order h, then $3 \leq h \leq p$. By (4.1) and Lemma 2.4, its star count is at most $\max\{h_r(q, p), g_r(q, p-1)\}$. Replace it by a graph attaining this maximum. Since every star lies in a component, these operations do not decrease $N(S_r, G)$. Each resulting component is either a clique of order at most p or a graph $G(q, p-1)$ with $q \geq p+1$. The detour order remains p. Indeed, a component originally containing a p-vertex path becomes K_p if its order is p, and otherwise is replaced by a graph of detour order p; all other replacements have detour order at most p.

If every component is complete, Lemma 2.2(iii), applied to $g(x) = x\binom{x-1}{r-1} = r\binom{x}{r}$ with cap p, yields the bound $h_r(n, p)$. Thus suppose that a G-component is present. We show that all components can be merged into one G-component without decreasing the star count. Let $t = \lfloor (p-1)/2 \rfloor$ and $\eta = p - 2t \in \{1, 2\}$.

First, consider two components $G(a, p-1)$ and $G(b, p-1)$, where $a, b \geq p+1$. Let $\Delta = \binom{a+b-1}{r-1} - \binom{a-1}{r-1} - \binom{b-1}{r-1}$ and $L = \binom{p-1}{r-2}$. By Vandermonde's identity,

$$\Delta = \binom{a-1}{r-2} + \sum_{i=1}^{r-2}\binom{a}{i}\binom{b-1}{r-1-i} \geq a\binom{b-1}{r-2} \geq (p-1)L.$$

Replacing these components by $G(a+b, p-1)$ changes the star count by

$$\begin{aligned} t\Delta + (p-t)\binom{t}{r-1} - \eta\binom{p-1-t}{r-1} &\geq \left(t - \frac{\eta}{r-1}\right)(p-1)L + (p-t)\binom{t}{r-1} \\ &\geq 0. \end{aligned}$$

Here we used $\binom{p-1-t}{r-1} \le \binom{p-1}{r-1} \le \frac{p-1}{r-1}L$ and $t \ge 1$, $\eta \le 2$, $r \ge 3$. Repeating this operation leaves exactly one G-component.

Next, replace $G(a, p-1)$ and a complete component K_b, where $a \ge p+1$ and $1 \le b \le p$, by $G(a+b, p-1)$. The change in the star count is

$$t\sum_{j=0}^{b-1}\binom{a-1+j}{r-2} + b\binom{t}{r-1} - b\binom{b-1}{r-1} \ge b\left[t\binom{p-1}{r-2} + \binom{t}{r-1} - \binom{p-1}{r-1}\right] \ge 0.$$

For the last inequality, note that $t \ge (p-2)/2$ and $r \ge 3$ imply $t(r-1) \ge p-2 \ge p-r+1$, so $t\binom{p-1}{r-2} \ge \binom{p-1}{r-1}$. By Proposition 2.1, every resulting G-component has detour order p. Merging all remaining complete components therefore produces $G(n, p-1)$ and proves the bound $g_r(n, p-1)$ in this case. The two cases prove the upper bound. The constructions show that it is sharp. □

5 Applications

The main theorems and Lemma 2.4 yield the following two corollaries.

Corollary 5.1. *Let $n \ge 4$ and $r \ge 3$. The maximum number of copies of S_r in an n-vertex non-Hamiltonian graph is*

$$\max\{f_r(n, n-1), g_r(n, n-1)\}.$$

Proof. Let G be an n-vertex non-Hamiltonian graph. If G contains a cycle, then $3 \le c(G) \le n-1$, and the upper bound follows from Theorem 1.4 and Lemma 2.4. If G is a forest, add edges between its components to obtain a tree, and then join the endpoints of a three-vertex path. The resulting graph has a unique cycle of length 3 and contains at least as many stars as G. Since $3 \le n-1$, the same upper bound applies. Finally, $F(n, n-1)$ and $G(n, n-1)$ are non-Hamiltonian and attain the two candidate values. □

Corollary 5.2. *Let $n \ge 4$ and $r \ge 3$. The maximum number of copies of S_r in an n-vertex non-traceable graph is*

$$\max\{h_r(n, n-1), g_r(n, n-2)\}.$$

Proof. Let G be an n-vertex non-traceable graph. If $p(G) \ge 3$, then $p(G) \le n-1$, and the upper bound follows from Theorem 1.5 and Lemma 2.4. If $p(G) \le 2$, then $\Delta(G) \le 1$, so G contains no S_r. The graphs $H(n, n-1)$ and $G(n, n-2)$ are non-traceable and attain the two candidate values. □

Acknowledgements

Leilei Zhang's work was supported by the National Natural Science Foundation of China (Grant Nos. 12601664, 12571365), JSPS KAKENHI (Grant No. 25KF0036), the Hubei Provincial Natural Science Foundation of China (Grant No. 2025AFB309), the China Postdoctoral Science Foundation (Grant No. 2025M773113), the Fundamental Research Funds for the Central Universities, Central China Normal University (Grant No. CCNU24XJ026).

Declarations

Conflict of interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability. Data sharing not applicable to this paper as no datasets were generated or analysed during the current study.

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